

13.12 Equivalent Expressions Using Properties of Operations

Lesson Introduction

 Benchmark	MA.7.AR.1.2 Determine whether two linear expressions are equivalent.		 Learning Targets	- I can use properties of operations to write equivalent forms of an algebraic expression. - I can determine whether two linear expressions are equivalent.
 Benchmark Coherence	Building On MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		Working Toward MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	How can you prove linear expressions are equivalent using properties of operations?			
 Lesson Narrative	Students extend the use of properties from numeric to algebraic expressions. Students begin by refining their working definitions of commutative, associative, and distributive properties in the context of numeric expressions before exploring what those properties look like through the lens of algebraic expressions. This lesson includes more spiraled practice on fluency with fractions.			
 Component Summary	13.12.1 Warm-Up (~5 min.)	Fact or Fiction? equivalent expressions (<i>SE p. 327</i>)	13.12.4 Guided Practice (10 min.)	Write equivalent algebraic expressions using the properties of operations (<i>SE p. 330</i>)
	13.12.2 Refresher (10 min.)	Reviewing associative, commutative, and distributive properties of numeric expressions to apply to algebraic expressions (<i>SE p. 328</i>)	13.12.5 Your Turn! (10 min.)	Students choose properties to rewrite equivalent algebraic expressions (<i>SE p. 330</i>)
	13.12.3 Guided Instruction (10 min.)	Extending properties from numeric to algebraic expressions (<i>SE p. 329</i>)	13.12.6 Wrap-Up (~5 min.)	Complete a self-assessment of ability to use properties of operations to write equivalent forms of algebraic expressions (<i>SE p. 331</i>)
 Resources	Glossary		Materials	
	Additional Terms: - Associative property - Distributive property - Commutative property		(Optional) Note cards (Optional) Chart paper (Optional) Markers	
			Additional Preparation Consider a Gallery Walk with chart paper for 13.12.2 Refresher.	

 Teacher Tips	Common Misconceptions - When rewriting expressions using the distributive property, students may forget to distribute to all terms. - Students may add instead of multiply coefficient and substituted values, like $3x \rightarrow 3(2) \rightarrow 5$, or fail to use parentheses and interpret this as 32.	Things to Consider - Similarly, for variable expressions, encourage students to substitute values into the expressions and evaluate each expression to verify that the expressions are equivalent. - Colors may be helpful when substituting values in for the variables. - Determine if your students need the support of all or part of the 13.12.2 Refresher.
	Literacy and Language Routines Think-Pair-Share In 13.12.4 Guided Practice, students create equivalent expressions using properties of operations. Prior to independently recording the properties, ask students to engage in a Think-Pair-Share to swap their expressions with a partner and determine the properties their partners used.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection After reviewing the associative, commutative, and distributive properties, students have multiple opportunities to apply, extend, and practice the properties. During partner work, encourage students to reflect on their own choice of properties to rewrite the expressions and compare that to those of their partner.
	 Differentiate Instruction	
13.12.5 Your Turn! offers student choice and multiple pathways as students apply any of the properties to different expressions. Consider a Four Corners activity in which each property is assigned a corner. Students must find a classmate from a different corner, show their work, and verify the work of another. Alternatively, students can hold up a number from one to three, with each number representing which property they applied. Students then must pair up with a classmate of the same number to verify their answers before finding a new partner with a different number and comparing their work.		
Lesson Notes:		

13.12.1 Warm-Up: Fact or Fiction?

Component Overview: Students recall the properties of operations to determine whether the numeric expression on the left side of an equation is equivalent to the numeric expression on the right side of the equation. To verify that numeric expressions are equivalent, encourage students to evaluate the expressions and review quickly as a class before moving on to 13.12.2 Refresher.



13.12.1 Warm-Up: Fact or Fiction?

- Determine whether each equation below is fact (true) or fiction (false).

Equation	Fact or Fiction
$(30 + 8) \cdot 9 = 270 + 72$	☐ Fact ☐ Fiction
$4 + 5(9 - 15) = 9(9 - 15)$	☐ Fact ☐ Fiction
$2 \cdot (7 + 8) = (2 \cdot 7) + 8$	☐ Fact ☐ Fiction
$(5 + 4) - 10 = 5 + (4 - 10)$	☐ Fact ☐ Fiction



Consider prompting questions to compare methods of evaluation and properties.

- Do you have to evaluate (or simplify) to determine if both sides of the equation are equivalent?
- Are the sides of each equation equal or equivalent?

13.12.2 Refresher: Rewriting Expressions Using Properties of Operations

Component Overview: Students apply the associative, commutative, and distributive properties to rewrite and evaluate expressions, connecting prior knowledge with numeric expressions to algebraic expressions. Then students work with a partner to discuss why the associative and commutative properties don't hold for all operations and create examples to prove their findings.



13.12.2 Refresher: Rewriting Expressions Using Properties of Operations

1. Fill in the missing blanks for each property.

- Using the **commutative property of addition**, $3 + 12$ can be rewritten as $\underline{12 + 3}$. Both expressions have a sum of $\underline{15}$.
- Using the **commutative property of addition**, $4x + 5$ can be rewritten as $\underline{5 + 4x}$. Both expressions will have the same sum regardless of the value of x .
- Using the **commutative property of multiplication**, $(4)(-2)$ can be rewritten as $\underline{(-2)(4)}$. Both expressions have a product of $\underline{-8}$.
- Using the **commutative property of multiplication**, $(-3y)(0.5)$ can be rewritten as $\underline{(0.5)(-3y)}$. Both expressions will have the same product regardless of the value of y .
- Using the **associative property of addition**, $-5 + (6 + 1)$ can be rewritten as $\underline{(-5 + 6) + 1}$. Both sums are equal to $\underline{2}$.
- Using the **associative property of addition**, $9x + (-2x + 5)$ can be rewritten as $\underline{(9x + -2x) + 5}$. Both expressions will have the same sum regardless of the value of x .
- Using the **associative property of multiplication**, $2 \cdot (7 \cdot 3)$ can be rewritten as $\underline{(2 \cdot 7) \cdot 3}$. Both products are equal to $\underline{42}$.
- Using the **associative property of multiplication**, $(-6)(4y)$ can be rewritten as $\underline{(-6 \cdot 4)y}$. Both expressions will have the same product regardless of the value of y .



Data-Driven Instruction

Allow data and informal observations from previous lessons in this unit to drive decision-making.

Small Groups: Utilize a teacher-led small group if you have a handful of students who need additional support.

Gallery Walk: Create three sets of all three properties (nine total so each poster only has three to four students at one time) on chart paper. Students must visit each property and write at least one example of that property. After they write one example for each property, they then must find a peer's examples and verify by writing a check mark if correct.

Utilize independent practice if students need time meaningfully engaging in producing examples. These are best utilized in context, so prioritize question 2 more than question 1.

13.12.2 Refresher: Rewriting Expressions Using Properties of Operations (continued)

- i. Using the **distributive property of multiplication over addition**, $5(6 + 2)$ can be rewritten as $(5 \cdot 6) + (5 \cdot 2)$. Both expressions have a value of 40.
 - j. Using the **distributive property of multiplication over addition**, $7(3y + 8)$ can be rewritten as $(7 \cdot 3y) + (7 \cdot 8)$ or $21y + 56$. Both expressions will have the same product regardless of the value of y .
2. Discuss with your partner why the associative property and the commutative property do not hold true for subtraction and division. Give an example of each to show mathematically that these properties do not apply to subtraction and division.
The associative property does not hold true for subtraction. For example: $(10 - 5) - 2 = 3$ is not equal to $10 - (5 - 2) = 7$.

The associative property does not hold true for division. For example: $(50 \div 5) \div 2 = 5$ is not equal to $50 \div (5 \div 2) = 20$.

The commutative property does not hold true for subtraction. For example: $10 - 5 = 5$ is not equal to $5 - 10 = -5$.

The commutative property does not hold true for division. For example: $10 \div 5 = 2$ is not equal to $5 \div 10 = \frac{1}{2}$.
 3. Jay gave the example $6 + ^{-}11$ is equal to $^{-}11 + 6$ to show that subtraction is commutative. Explain to Jay why his example does not show that subtraction is commutative.
Jay's problem shows that the commutative property holds for addition.



Consider providing sentence starters for students who may struggle with written explanations. Alternatively, allow students to explain their responses orally to provide an additional entry point.

13.12.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Component Overview: Students apply the associative, commutative, and distributive properties to determine if algebraic expressions are equivalent.



13.12.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Algebraic expressions are considered equivalent if when evaluating the expression by substituting values in for variables, the resulting values are the same.

Algebraic expressions are also considered equivalent when one expression is rewritten using one or more of the properties of operations.

1. Complete each statement.

- a. The ☐ associative
☐ commutative
☒ distributive property can be used to

state that $5(x + 2) - 3x$ is equivalent to $5x + 10 - 3x$.

- b. The ☐ associative
☒ commutative
☐ distributive property can be used to

state that $\frac{2}{3}x + \left(\frac{3}{8} + \frac{5}{6}x\right)$ is equivalent to $\frac{2}{3}x + \left(\frac{5}{6}x + \frac{3}{8}\right)$.

2. Which properties can be used to show that $2(x - 4) + 5(3 + 6x)$ is equivalent to $2x + 30x - 8 + 15$?

The distributive property of multiplication over addition and over subtraction were used to obtain $2x - 8 + 15 + 30x$. Then, the commutative property was used to rearrange the terms and obtain $2x + 30x - 8 + 15$.



How can you use properties of operations to determine whether algebraic expressions are equivalent?



For students who may finish early, encourage them to rewrite the expression in question 2 in its simplest form and explain how they know it is in simplest form.

13.12.4 Guided Practice: Rewriting Expressions Using Properties of Operations

Component Overview: Students rewrite equivalent algebraic expressions using the properties of operations.



13.12.4 Guided Practice: Rewriting Expressions Using Properties of Operations

1. Write an equivalent expression using a property of operations. State the property used.

Answers will vary.

Expression	Equivalent Expression	Property Used
$-6(2 - 3x) + 1 - \frac{x}{2}$	$= -12 + 18x + 1 - \frac{1}{2}x$ $= 18x - \frac{1}{2}x - 12 + 1$ $= 17\frac{1}{2}x - 11$	Distributive & Commutative Property
$12x + 10 + 4x - 17$	$= 16x - 7$	Commutative Property
$5x + (3x + 9) + 2$	$= 8x + 11$	Associative Property

2. Several expressions are given below. Determine which expressions are equivalent using properties of operations.

$35x + (7x + 56)$	$5(8x + 7)$	$35x + 56$
$(8x + 7) \cdot 5$	$7(5x + 8)$	$(35x + 7x) + 56$

Complete the table by writing the expressions that are equivalent based on the property used.

Equivalent Expressions		Property Used
$35x + (7x + 56)$	$(35x + 7x) + 56$	Associative Property
$5(8x + 7)$	$(8x + 7) \cdot 5$	Commutative Property
$35x + 56$	$7(5x + 8)$	Distributive Property



Think-Pair-Share

Encourage mathematical discourse by implementing a Think-Pair-Share to position students as property experts in creating, verifying, and justifying responses.



Challenge students to write an equivalent expression that uses all three properties of operations (the associative, commutative, and distributive properties).

13.12.5 Your Turn!

Component Overview: Students write equivalent algebraic expressions using the properties of operations. There is more than one correct equivalent expression.



13.12.5 Your Turn!

Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

Answers will vary.

1. $10x + \frac{2}{5}(-30 + 15x)$
 $= 10x + -12 + 6x$
Distributive property of multiplication over addition
2. $(18x + 15) + 4x - 17$
 $= (15 + 18x) + 4x - 17$
Commutative property of addition
3. $(-\frac{1}{5}x + 6) \cdot 5$
 $= 5 \cdot (-\frac{1}{5}x + 6)$
Commutative property of multiplication
4. $\frac{5}{4}n + \frac{7}{3} - \frac{11}{2}n - 12\frac{1}{3}$
 $= \frac{5}{4}n + -\frac{11}{2}n + \frac{7}{3} + -12\frac{1}{3}$
Commutative property of addition
5. $-3(\frac{1}{2}x + 4) - \frac{3}{5}(-1 + 10x)$
 $= \frac{-3}{2}x - 12 - \frac{3}{5}(-1 + 10x)$
Distributive property of multiplication over addition



There is more than one correct equivalent expression. Ask students to share and justify how they used properties to write an equivalent expression and consider having students evaluate the examples they have come up with as a class to encourage analysis. Correct student responses could be used to create a Card Sort as a review activity before the unit assessment.



Consider providing a specific property to use to rewrite each expression. This will encourage students to practice all three properties and may provide an entry point for students who may struggle getting started on their own.

13.12.6 Wrap-Up: Ticket Out the Door

Component Overview: Students use properties of operations to determine if two linear expressions are equivalent.



13.12.6 Wrap-Up: Ticket Out the Door

Use properties of operations to determine if the expressions are equivalent.

1. $\frac{4}{5}(x - 1) - \frac{1}{8}(2x + 1)$ and $\frac{11}{20}x - \frac{37}{40}$

The expressions are equivalent.

$$\begin{aligned} & \frac{4}{5}x - \frac{4}{5} - \frac{2}{8}x - \frac{1}{8} \\ &= \left(\frac{4}{5}x - \frac{2}{8}x\right) + \left(-\frac{4}{5} - \frac{1}{8}\right) \\ &= \left(\frac{11}{20}x\right) + \left(-\frac{37}{40}\right) \\ &= \frac{11}{20}x - \frac{37}{40} \end{aligned}$$

2. $1\frac{3}{10}x + \frac{1}{5}x - \frac{1}{10} - x - \frac{1}{10}$ and $1\frac{1}{10} - \frac{4}{5}x$

The expressions are not equivalent.

$$\begin{aligned} & \left(1\frac{3}{10}x + \frac{1}{5}x - x\right) + \left(-\frac{1}{10} - \frac{1}{10}\right) \\ &= \left(\frac{3}{10}x + \frac{2}{10}x\right) + \left(-\frac{2}{10}\right) \\ &= \left(\frac{5}{10}x\right) + \left(-\frac{2}{10}\right) \\ &= \frac{1}{2}x - \frac{1}{5} \end{aligned}$$



Consider prompting students to provide more information to justify their reasoning:

- Students can show their work and label each step of the work with a property.
- Prompt students to show their work and write a one sentence summary describing which property they used to justify equivalence.